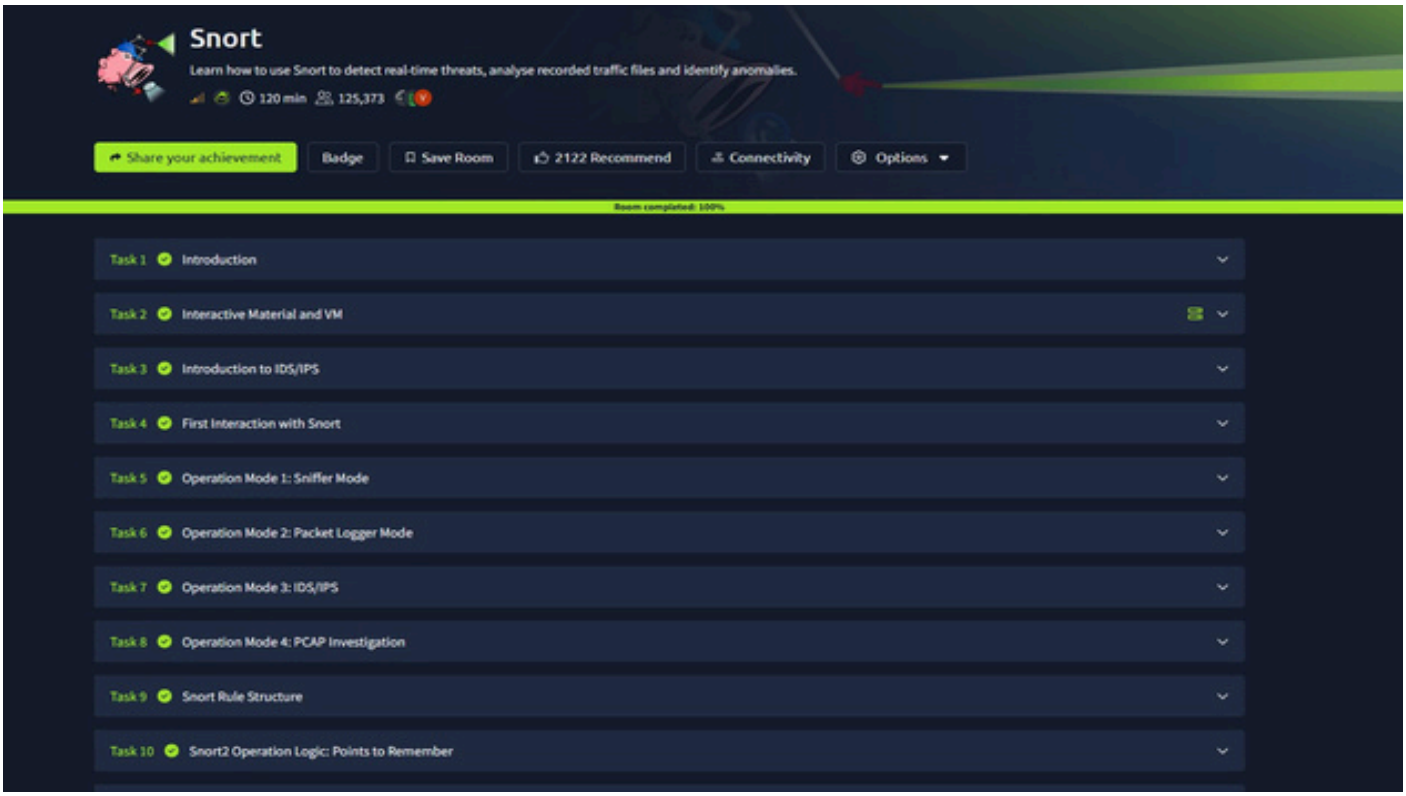
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CRYPTOGRAPHY

SNORT IDS

Aim:
To demonstrate Intrusion Detection System (IDS) using snort tool.
Algorithm:
1. Download and extract the latest version of daq and snort
2. Install development packages - libpcap and pcre.
3. Install daq and then followed by snort.
4. Verify the installation is correct.
5. Create the configuration file, rule file and log file directory
6. Create snort.conf and icmp.rules files
7. Execute snort from the command line
8. Ping to yahoo website from another terminal
9. Watch the alert messages in the log files
Output:



- **Sniffer Mode**: Read and prompt IP packets in the console application.
- **Packet Logger Mode**: Log all IP packets (inbound and outbound) that visit the network.
- **NIDS (Network Intrusion Detection System) and NIPS (Network Intrusion Prevention System) Modes**: Log/drop the packets deemed malicious according to the user-defined rules.

Answer the questions below

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS — ✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS — ✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS — ✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS — ✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA — ✓ Correct Answer

According to the official description of the snort, what kind of NIPS is it?

full-blown — ✓ Correct Answer

NBA training period is also known as ...

baselining — ✓ Correct Answer

```
1
2 # ----------------
3 # LOCAL RULES
4 # ----------------
5 # This file intentionally does not come with signatures.  Put your local
6 # additions here.
7
8 alert icmp any any <> any any (msg: "ICMP Packet Found"; sid: 1000001; rev:1;)
9
10
```

Note that some default rules are activated with the Snort instance. These rules are deactivated to manage your rules and improve your exercise experience. For further information, please refer to the TASK-10 or Snort manual.

By this point, we have covered the primary structure of the Snort rules. Understanding and practicing the fundamentals is suggested before creating advanced rules and using additional options.

Wow! We have covered the fundamentals of the Snort rules! Now, use the attached VM and navigate to the `Task-Exercises/Exercise-Files/TASK-9` folder to answer the questions! Note that you can use the following command to create logs in the current directory: -l

Answer the questions below

Use "task9.pcap". Write a rule to filter IP ID "35369" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l . -r task9.pcap"

TIMESTAMP REQUEST — ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Create a rule to filter packets with **Syn** flag and run it against the given pcap file. What is the number of detected packets?

1 — ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Write a rule to filter packets with **Push-Ack** flags and run it against the given pcap file. What is the number of detected packets?

216 — ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Create a rule to filter **UDP** packets with the same source and destination IP and run it against the given pcap file. What is the

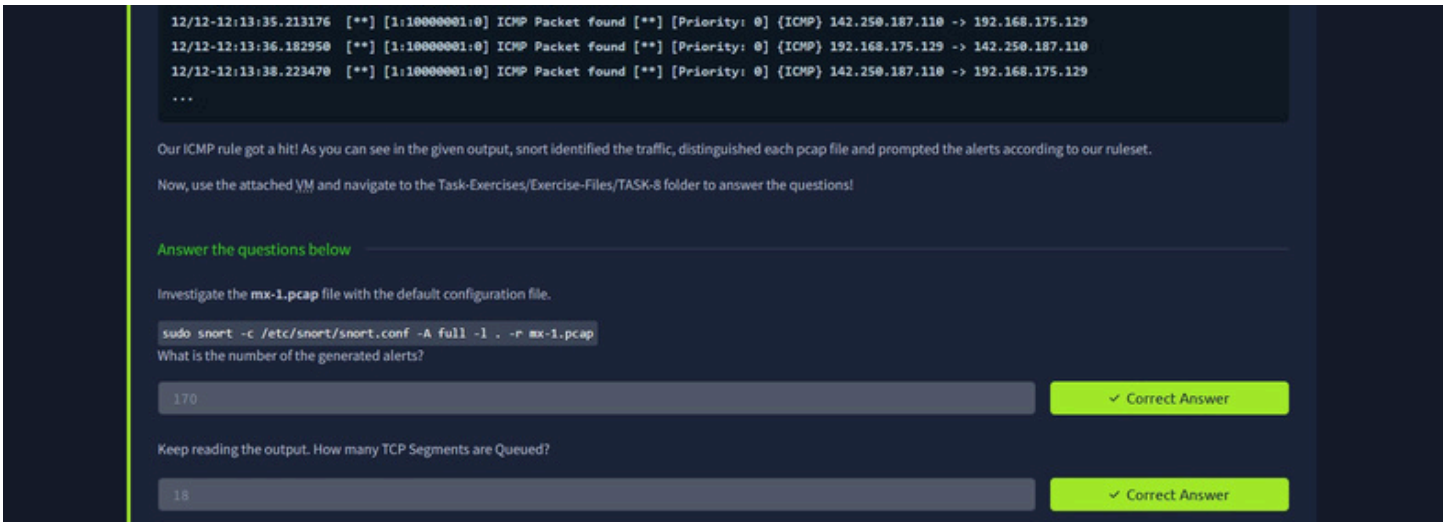
Let's investigate multiple pcaps, distinguish each one, and see what will happen. `sudo snort -c /etc/snort/snort.conf -q --pcap-list="icmp-test.pcap http2.pcap" -A console --pcap-show`

investigating multiple pcap files wth pcap info

```
user@ubuntu$ sudo snort -c /etc/snort/snort.conf -q --pcap-list="icmp-test.pcap http2.pcap" -A console --pcap-show

Reading network traffic from "icmp-test.pcap" with snaplen = 1514
12/12-12:13:29.167955  [**] [1:10000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.129 -> 142.250.187.110
12/12-12:13:29.200543  [**] [1:10000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 142.250.187.110 -> 192.168.175.129
12/12-12:13:30.169785  [**] [1:10000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.129 -> 142.250.187.110
...

Reading network traffic from "http2.pcap" with snaplen = 1514
```



Result: Thus, the Intrusion Detection System (IDS) has been successfully demonstrated using snort.